Does Confidence-Gating Work?
Empirical Evaluation of a Core AI Oversight Assumption

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Technical Report / Preprint · July 2026
https://github.com/Rushhaabhhh/Calibrated-Oversight

Abstract

Three 2026 governance frameworks—Singapore's Model AI Governance Framework for Agentic AI, the U.S. NIST/NCCoE concept paper on agent identity and authorisation, and the EU AI Act (Arts. 14–15)—all require "meaningful human oversight" of autonomous AI agents. A common implementation assumption is that agents can flag low-confidence decisions for human review. We empirically test this assumption, to our knowledge among the first such evaluations in the governance context. Verbalized confidence is elicited from two models (Llama 3.3 70B, Gemini 2.5 Flash) across four domains (N=2,004 decisions). On structured question-answering tasks the assumption holds: flagging the least-confident 10% finds errors at 1.7–2.9× the base rate. On open-ended software-engineering judgment (SWE-bench) it fails: lift $\approx$ 1.0 for both models, and stated confidence occupies such a narrow band (σ<0.06) that fixed thresholds are ineffective. In a preliminary comparison on SWE-bench (n=110), self-consistency—sampling the model three times and using answer agreement—raises lift to 2.06, suggesting that multi-sample strategies warrant further investigation as oversight mechanisms for open-ended agent tasks. A working HMAC-signed audit log implements the tamper-evidence requirement these frameworks also mandate. The governance implication is specific: confidence-gating requirements that do not distinguish by task type risk failing on the open-ended judgments that characterise many deployed agent settings.

1. Introduction

Autonomous AI agents are being deployed in software engineering, finance, and operations faster than the oversight mechanisms to govern them. Three major 2026 governance instruments independently formalise two requirements: agents must support *meaningful human oversight* and produce *tamper-evident audit trails* (IMDA, 2026; NIST/NCCoE, 2026; EU, 2024).

A natural and widely assumed implementation of the oversight requirement is **confidence-gating**: the agent flags any decision whose stated confidence falls below a threshold, directing limited human attention to the decisions most likely to be wrong (Galileo, 2026; Parasuraman & Manzey, 2010). This assumption is embedded in practitioner guides, evaluation pipelines, and proposed technical standards for agent deployment.

What has not been measured, in the context these frameworks care about, is whether the assumption holds. Do current models produce confidence estimates that actually identify their errors? Do those estimates work on the *kinds of tasks* agents perform in practice—open-ended, multi-step judgments—or only on structured question-answering?

This paper provides that measurement. We make three contributions:

(1) Empirical test of the confidence-gating assumption across four task types and two models, with proper statistics. On three structured QA domains the assumption holds. On software-engineering judgment it does not.

(2) Self-consistency as a stronger alternative for agent tasks. Where verbal confidence yields lift $\approx$ 1.0 on SWE-bench, vote-based self-consistency over three samples yields lift = 2.06 on the same items.

(3) A working audit-log prototype that implements the tamper-evidence requirement in the Python standard library, with a field-by-field mapping to regulatory clauses.

2. Why This Test Matters

The governance frameworks do not run experiments. IMDA draw on input from 50+ organisations but provide principles, not measurements. NIST's approach is "explicitly facilitative rather than prescriptive" (CSA, 2026): the agency identifies gaps and co-invests with industry but has not benchmarked specific oversight mechanisms. The EU AI Act (Art. 14(4)(b)) requires systems to support awareness of automation bias (Parasuraman & Manzey, 2010) and (Art. 14(4)(d)) to enable human override, but specifies neither how to measure confidence nor what threshold to use.

The research community has established that LLMs are systematically overconfident (Guo et al., 2017; Tian et al., 2023; Xiong et al., 2024) and that verbalized confidence is often better-calibrated than token probabilities for RLHF-tuned models (Tian et al., 2023; Kadavath et al., 2022). Concurrent work on coding agents finds severe overconfidence across pre-, mid-, and post-execution stages (Zhang et al., 2026). Selective prediction (Geifman & El-Yaniv, 2017) and its variants (Manakul et al., 2023; Kuhn et al., 2023) offer a richer framework for assessing when a model can and cannot be trusted—but these methods have not been evaluated against governance requirements or on the task types agents commonly face.

What has not been measured is whether confidence provides a *useful oversight trigger*—not just a calibrated number, but a number that actually finds errors—and whether this varies by task type in ways that matter for how governance requirements should be written.

Table 1: Proposed audit-record fields and the requirements each satisfies.

Field	Requirement
`step_id, timestamp`	IMDA: lifecycle monitoring
`agent_id`	NCCoE: attribute actions to non-human entity
`action, input_hash`	IMDA: record agent action and intent; hash limits data exposure
`confidence`	Art. 14(4)(b): automation-bias awareness
`oversight_flagged`	Art. 14(4)(d): trigger for human override
`prev_hash`	NCCoE: tamper-evidence via chain integrity
`signature`	NCCoE: non-repudiation

3. Regulatory Requirements and Audit Record

Table 1 maps each field of the proposed per-decision audit record to the specific regulatory clause it satisfies. This is the paper's governance engineering contribution: translating three sets of prose requirements into one concrete data structure that a compliance team could build against.

The `confidence` and `oversight_flagged` fields are useful only if the confidence signal is informative. The rest of this paper tests that.

4. Method

4.1 Tasks

We evaluate across four domains spanning qualitatively different task structures. **SWE-bench Verified** (Jimenez et al., 2024) contains genuine GitHub issues; models estimate $p(\text{fix}{>}15\,\text{min})$ against human difficulty labels. This is an *external difficulty judgment*: the model predicts a human annotation, not its own correctness. **GSM8K** (Cobbe et al., 2021) provides math word problems with exact numeric answers. **MMLU** (Hendrycks et al., 2021) samples academic MCQs across 20 subjects. **TruthfulQA** (Lin et al., 2022) provides adversarial factual questions. On the three QA domains, models assess p(own answer is correct)—a genuinely introspective quantity. This construct difference matters: the SWE-bench null result may partly reflect that models are poor difficulty estimators, not only poor introspectors.

4.2 Models and elicitation

Llama 3.3 70B (Groq) and Gemini 2.5 Flash (Google AI Studio) are evaluated via public APIs; no fine-tuning is performed. Each item receives a single probability elicitation (0–100). An earlier design eliciting a binary prediction plus a separate confidence rating led to uniformly high confidence regardless of direction, silently disabling the oversight gate.

For self-consistency (Wang et al., 2023), the same items are queried three times at temperature 0.3; the fraction voting for the majority answer serves as the confidence estimate. This is a simplified variant of SelfCheckGPT (Manakul et al., 2023), adapted for binary classification tasks.

4.3 Metrics

Oversight lift is the primary metric: the error rate among the least-confident 10% of decisions divided by the overall base error rate. Lift > 1 means low confidence reliably identifies errors; lift ≈ 1 means the gate adds no value over random selection. **AUC-RC** (Geifman & El-Yaniv, 2017) (area under the risk-coverage curve) measures whether sorting by confidence ranks decisions by correctness globally. Both metrics are reported with 2,000-sample bootstrap 95% CIs and permutation p-values for domain comparisons.

We also report ECE (Guo et al., 2017) and Brier score (Brier, 1950) before and after three post-hoc recalibration methods (temperature scaling, Platt scaling, isotonic regression), each evaluated five-fold out-of-sample. Confidence-in-own-answer standard deviation σ flags *confidence collapse* when $\sigma{<}0.10$.

5. Results

5.1 Confidence-gating works on QA, fails on agent tasks

Figure 1 shows the main result. On SWE-bench, both models produce lift ≈ 1: the least-confident 10% of decisions are only marginally more error-prone than average (Llama 1.03, Gemini 1.29). Flagging those items provides essentially no oversight value. On GSM8K, MMLU, and TruthfulQA, lift is substantially above 1 across both models: 2.86, 2.06, and 2.41 for Llama; 0.82[†], 1.91, and 1.67 for Gemini.

The domain dependence is statistically robust. The Llama 70B GSM8K–SWE-bench lift difference is $+1.83$ (95% CI $[+1.03, +2.63]$, permutation $p{<}0.001$); MMLU–SWE-bench is $+1.02$ ($[+0.14, +1.90]$, $p{=}0.014$); TruthfulQA–SWE-bench is $+1.38$ ($[+0.32, +2.45]$, $p{=}0.009$). A two-proportion z-test finds no significant accuracy difference between models on SWE-bench ($\Delta{=}{-}0.008$, $p{=}0.90$), ruling out model-specific effects.

[†] *Gemini's GSM8K lift of 0.82 reflects complete confidence collapse ($\sigma{=}0.000$): every item received 100% confidence, so the flagged decile has no different error rate from average.*

The reliability diagrams (Figure 3) confirm the calibration picture. On GSM8K, Llama achieves 69.2% accuracy but reports 98.6% mean confidence (ECE = 0.294). Gemini reports 100% confidence on every item ($\sigma{=}0.000$, ECE = 0.440). On MMLU and TruthfulQA both models are better calibrated.

The confidence distributions (Figure 4) explain why the gate fails: $\sigma{<}0.10$ in every condition. The distributions of correct and incorrect decisions overlap almost entirely on SWE-bench. No fixed threshold cleanly separates them; the same cutoff flags 8.7% of one model's decisions and 0% of another's—not because the models differ in reliability but because their distributions sit at different absolute levels.

5.2 Post-hoc calibration fixes aggregate error but not the gate

Figure 6 shows that post-hoc recalibration substantially reduces ECE. Platt scaling reduces Llama's SWE-bench ECE from 0.201 to 0.015 (five-fold out-of-sample). In every condition, at least one method reduces ECE by more than 50% relative. Aggregate calibration is technically recoverable without retraining (Guo et al., 2017; Tian et al., 2023).

However, this does not restore the oversight gate. Post-hoc calibration corrects the *aggregate* relationship between stated confidence and empirical accuracy, but it cannot manufacture per-decision discrimination absent from the raw signal. On SWE-bench, corrected confidence still does not rank decisions by correctness because the underlying signal is uniformly high.

5.3 Self-consistency is a stronger mechanism for agent tasks

Figure 2 shows the self-consistency comparison on SWE-bench and GSM8K (Llama 3.3 70B, n=110, 3 samples each). On SWE-bench, verbal confidence yields lift = 1.47 and AUC-RC = 0.669; self-consistency yields lift = 2.06 and AUC-RC = 0.623. Vote-fraction agreement is a substantially stronger error signal for this open-ended task. On GSM8K, verbal confidence is the stronger signal (lift = 2.86 vs. SC = 1.95), consistent with models being able to express genuine introspective uncertainty about arithmetic.

The likely mechanism: sampling probes the model's implicit probability distribution over answers, making consistency a richer signal than a verbalised number produced in a single forward pass (Manakul et al., 2023; Kuhn et al., 2023). For governance frameworks requiring an oversight trigger on agent tasks, verbal confidence is insufficient; multi-sample agreement provides a meaningfully better signal at a 3× API cost per decision—acceptable for high-stakes items.

5.4 Miscalibration concentrates on easy tasks

Stratifying Llama 70B on SWE-bench by difficulty: on issues labelled "<15 min fix" the model is 30% accurate while reporting 81% mean confidence (gap +0.51). On 15-min-to-4-hour issues the gap is under 0.10. The model treats nearly all issues as hard, failing precisely where an overseer would least expect it. By repository, overconfidence concentrates on `sphinx-doc` (gap +0.31) and `sympy` (+0.23).

6. Implementing the Audit-Trail Requirement

We implement the Table 1 schema as an append-only, hash-chained, HMAC-signed log using only the Python standard library, following the tamper-evidence design of Crosby & Wallach (2009). Each entry stores `prev_hash` (SHA-256 of the previous entry's canonical JSON) and a `signature` (HMAC-SHA256 over the entry content). A `verify()` function re-walks the chain and localises the first failure.

Two insider-attack simulations are run on a 150-entry chain: flipping `oversight_flagged` from `true` to `false` (suppressing a review trigger), and rewriting a `confidence` value upward (faking certainty). Both are detected and localised to the exact tampered entry; the intact chain verifies cleanly.

This is a prototype. HMAC is symmetric; production requires Ed25519 and managed key rotation. What it demonstrates—append-chaining, per-entry signing, and tamper localisation—matches what the NIST/NCCoE concept paper asks for at the mechanism level.

7. Discussion

Why confidence fails on agent tasks. SWE-bench elicits $p(\text{fix}>15\,\text{min})$, an external difficulty prediction against a human annotation—not a self-assessment of the model's correctness. Even a perfectly introspective model might have no useful prior over fix duration. The QA tasks elicit $p(\text{own answer is correct})$, a genuinely introspective quantity. Current models appear capable of meaningful self-assessment on closed-ended reasoning but not on open-ended difficulty estimation. Both failures are governance-relevant; they call for different interventions.

Implications for framework design. A requirement that agents "flag low-confidence outputs" without specifying task type will be implemented with verbal confidence gates, which work on structured tasks and fail on open-ended agent judgments. The specific recommendation is: governance requirements for confidence-based oversight should (a) specify whether they intend introspective confidence or difficulty estimation, (b) require empirical validation that the chosen signal provides oversight lift > 1 on the target task distribution, and (c) consider multi-sample agreement as a stronger alternative for open-ended tasks.

Limitations. The study uses single-turn judgment proxies for multi-step agent behaviour. Only two model families were tested; reasoning models with chain-of-thought may have different self-assessment properties (Kadavath et al., 2022). Sample sizes of 110–250 produce CIs spanning 0.08–0.12 ECE units. The self-consistency comparison is preliminary (n=110, SWE-bench only). The audit log uses symmetric signing. The EU AI Act's operative timeline is under active revision.

8. Conclusion

Three 2026 governance frameworks require confidence-based human oversight of AI agents but do not test whether the mechanism works. We run that test. On structured QA tasks, verbalized confidence provides a useful oversight signal (lift up to 2.9). On software-engineering judgment—a representative open-ended agent task—it does not (lift ≈ 1.0). A preliminary self-consistency comparison (n=110) raises SWE-bench lift to 2.06. Governance frameworks that mandate confidence-gating without specifying task type, or without requiring empirical validation on the target distribution, risk failing silently on the open-ended judgments where deployed agents most commonly operate.

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Appendix: Figures

Note. Figures are placed in the appendix to allow the main text to run uninterrupted. All figures are referenced in the results section; readers are encouraged to view them alongside the corresponding subsection.

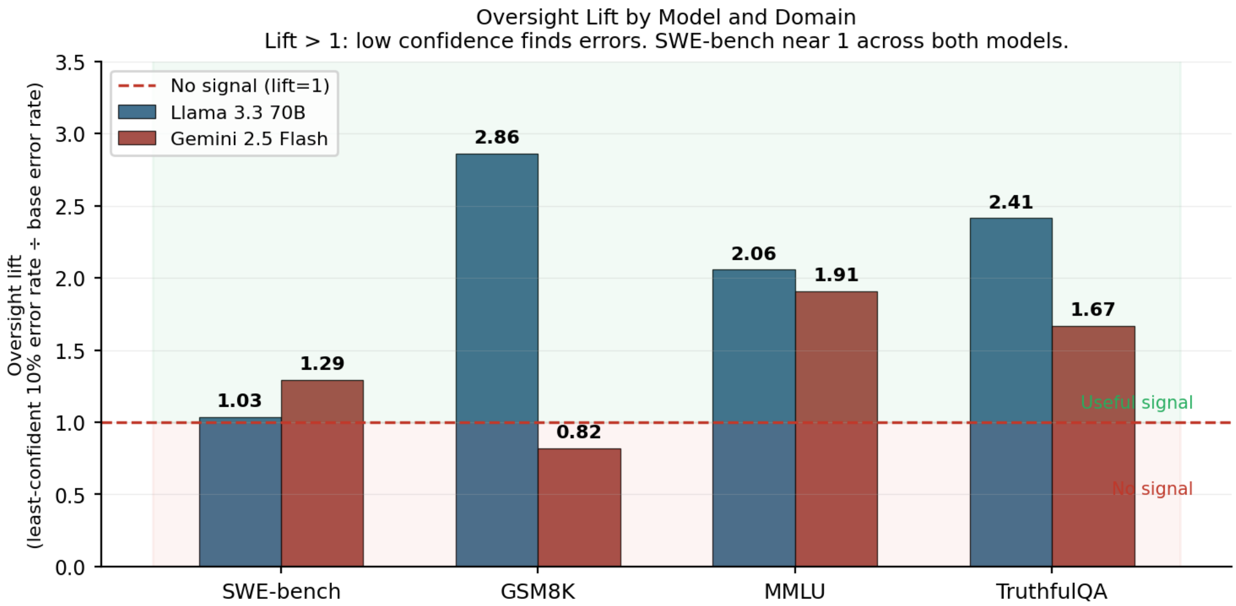

Figure 1: Fig. 1 — Oversight Lift by Model and Domain. Lift = error rate among the least-confident 10% ÷ overall base error rate. Lift > 1 (green region): low confidence reliably finds errors; lift ≈ 1 (red region): the gate adds no value. SWE-bench lift is near 1 for both models across every configuration tested. GSM8K, MMLU, and TruthfulQA lifts are 1.67–2.86. *This is the paper's primary result: confidence-gating works on structured QA but not on open-ended agent tasks.*

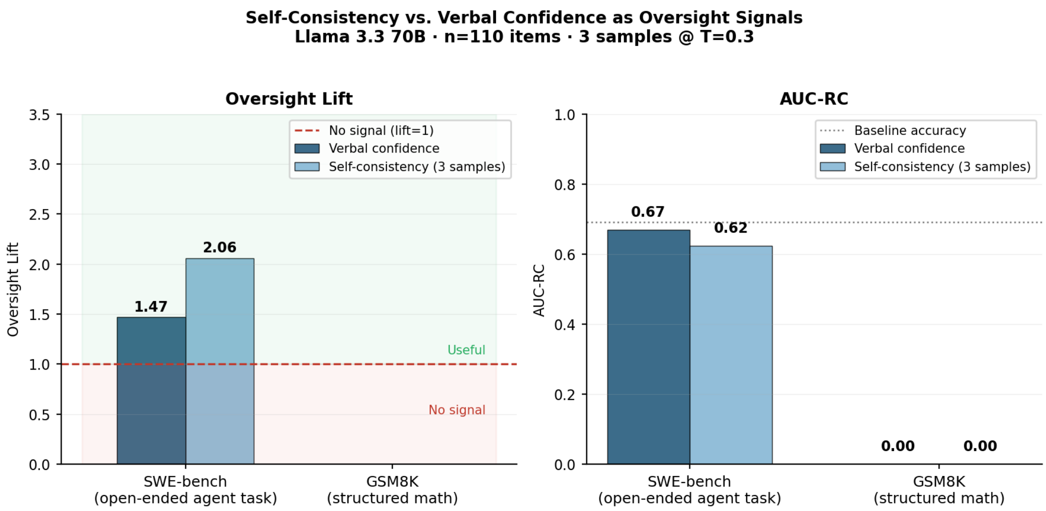

Figure 2: Fig. 2 — Self-Consistency vs. Verbal Confidence as Oversight Signals (Llama 3.3 70B, n=110). *Left*: oversight lift. *Right*: AUC-RC. On SWE-bench, self-consistency (3 samples, T=0.3) raises lift from 1.47 to 2.06—a 40% improvement on the task type where verbal confidence fails. On GSM8K, verbal confidence is the stronger signal (lift 2.86 vs. SC 1.95). Neither method dominates across task types, suggesting that the choice of uncertainty estimator should be matched to the task.

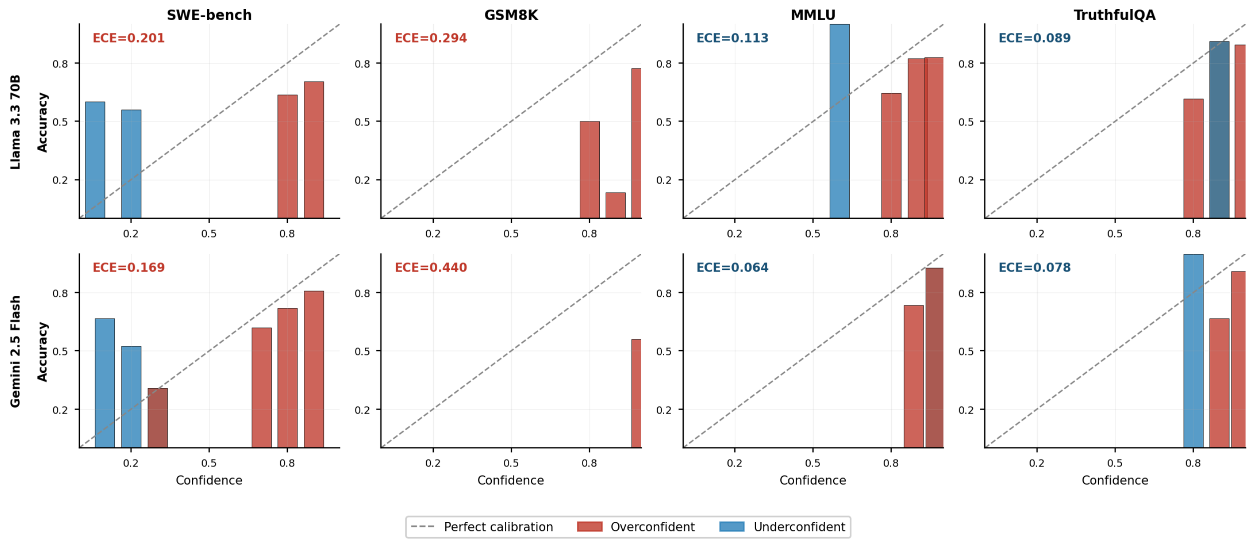

Figure 3: Fig. 3 — Reliability Diagrams (2 models × 4 domains). Bar height is empirical accuracy in each confidence bin; the dashed diagonal is perfect calibration. Red bars: overconfident (stated confidence exceeds accuracy). Blue: underconfident. GSM8K panels show systematic overconfidence for both models, with bars far above the diagonal. On MMLU and TruthfulQA, calibration is closer to the diagonal but bins are sparse due to confidence collapse.

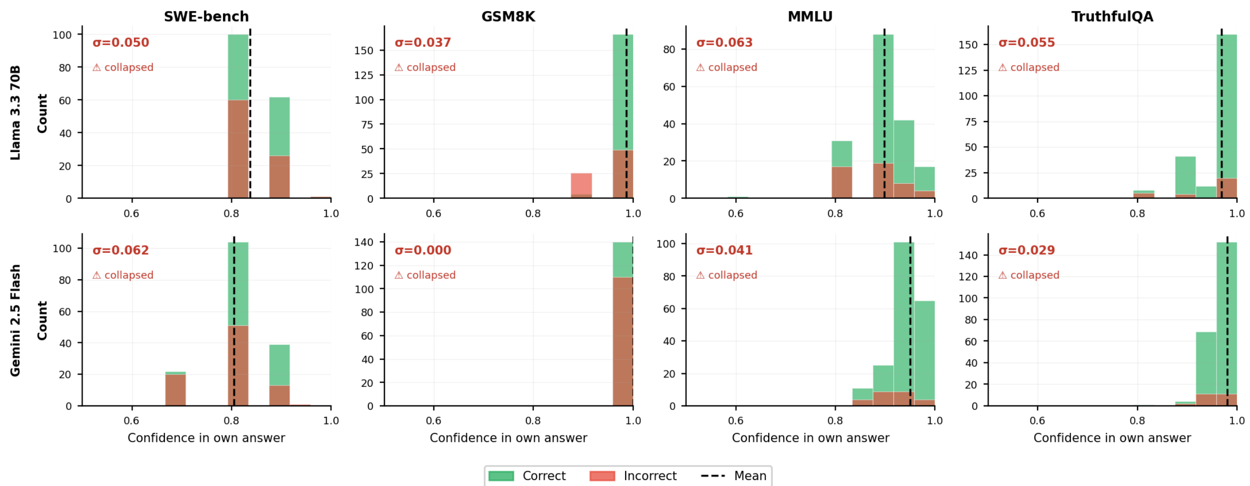

Figure 4: Fig. 4 — Confidence Distributions by Domain and Model. Histograms split by outcome (green: correct; red: incorrect). Every panel has $\sigma<0.10$ (labelled), indicating confidence collapse: the distributions of correct and incorrect decisions overlap almost entirely. Gemini's GSM8K column contains all 250 observations in a single bar at 1.0 (σ=0.000). Because the distributions are indistinguishable, no fixed threshold can reliably separate reliable from unreliable decisions.

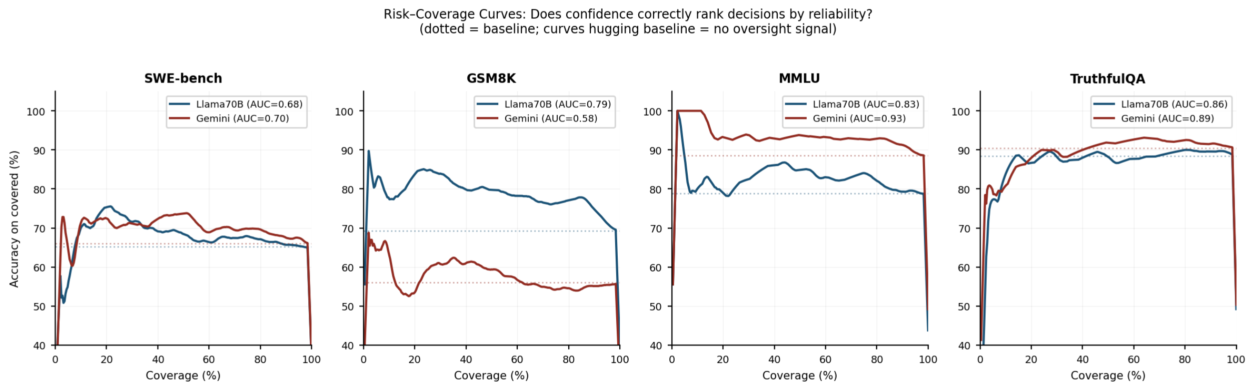

Figure 5: Fig. 5 — Risk-Coverage Curves (one panel per domain). The x-axis is the fraction of decisions handled autonomously (sorted most-confident first); the y-axis is accuracy on that covered set. Dotted horizontal lines show each model's baseline accuracy. Curves that track the baseline—as both models do on SWE-bench—indicate that confidence does not rank decisions by correctness. On GSM8K, MMLU, and TruthfulQA, curves are meaningfully above baseline at low coverage.

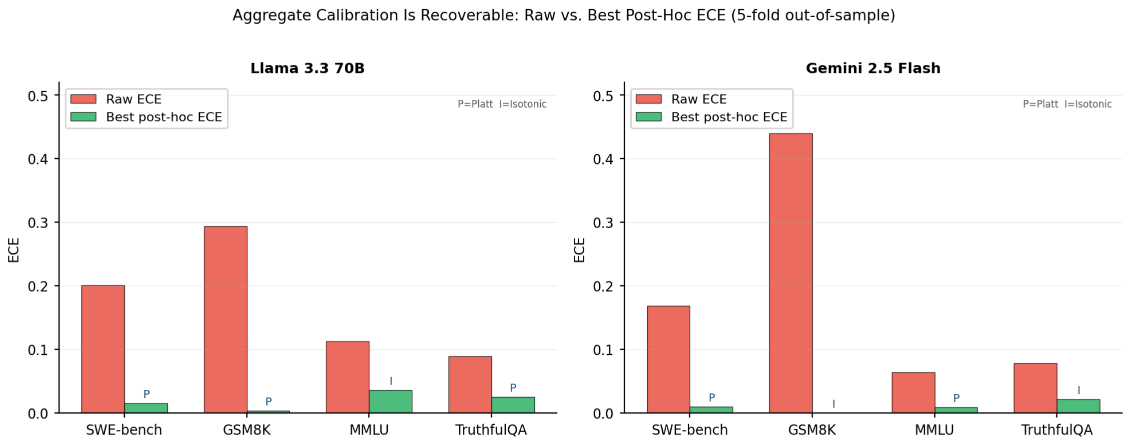

Figure 6: Fig. 6 — Aggregate Calibration Before and After Post-Hoc Correction. Raw ECE (red) vs. best five-fold out-of-sample post-hoc ECE (green). P=Platt scaling, I=Isotonic regression. Every condition achieves >50% relative ECE reduction. *Aggregate calibration is recoverable without retraining; per-decision discrimination is not.* The Gemini-GSM8K green bar reaches near-zero because isotonic regression maps the single confidence value (1.0 for all items) to the mean accuracy—ECE = 0 by construction, not by calibration.